

MNE-Python: EEG Power Spectrum

This example demonstrates basic frequency-domain analysis of EEG data using the `Spectrum` class in MNE-Python.

We will:

1. download a small public EEG recording;
2. load it as an MNE `Raw` object;
3. attach standard EEG sensor positions;
4. calculate the power spectral density (PSD);
5. visualize the spectrum;
6. visualize the spatial distribution of spectral power.

The example is inspired by the official MNE tutorial:

The `Spectrum` and `EpochsSpectrum` classes: frequency-domain data

Dataset

We use the EEG Motor Movement/Imagery Dataset distributed through PhysioNet and directly accessible through MNE.

The dataset contains 64-channel EEG recordings.

For this example we use:

- **Subject:** 1
- **Run:** 2
- **Condition:** resting-state EEG

Only one recording is downloaded.

✓ **General**

Filename(s)	S001R02.edf
MNE object type	RawEDF
Measurement date	2009-08-12 at 16:15:00 UTC
Participant	X
Experimenter	Unknown

✓ **Acquisition**

Duration	00:01:01 (HH:MM:SS)
Sampling frequency	160.00 Hz
Time points	9,760

✓ **Channels**

EEG	64
Head & sensor digitization	67 points

✓ **Filters**

Highpass	0.00 Hz
----------	---------

Recording information

The recording contains:

- **Channels:** 64
- **Sampling frequency:** 160.0 Hz
- **Duration:** 61.0 seconds

The data are represented by an MNE `Raw` object.

Power Spectral Density

Power spectral density describes how signal power is distributed across frequencies.

MNE provides `Raw.compute_psd()` for calculating a spectral representation directly from continuous EEG data.

Here we calculate frequencies between **1 and 45 Hz** using Welch's method.

✓ **General**

MNE object type	Spectrum
Measurement date	2009-08-12 at 16:15:00 UTC
Participant	X
Experimenter	Unknown

✓ **Acquisition**

Sampling frequency	160.00 Hz
--------------------	-----------

✓ **Channels**

EEG	64
Head & sensor digitization	67 points

✓ **Frequencies**

Data type	Power Spectrum
Computed from	Raw
Estimation method	welch
Frequency range	1.02 – 45.00 Hz

The resulting spectral representation contains:

- **EEG channels:** 64
- **Frequency bins:** 564
- **Frequency range:** 1.02–45.00 Hz

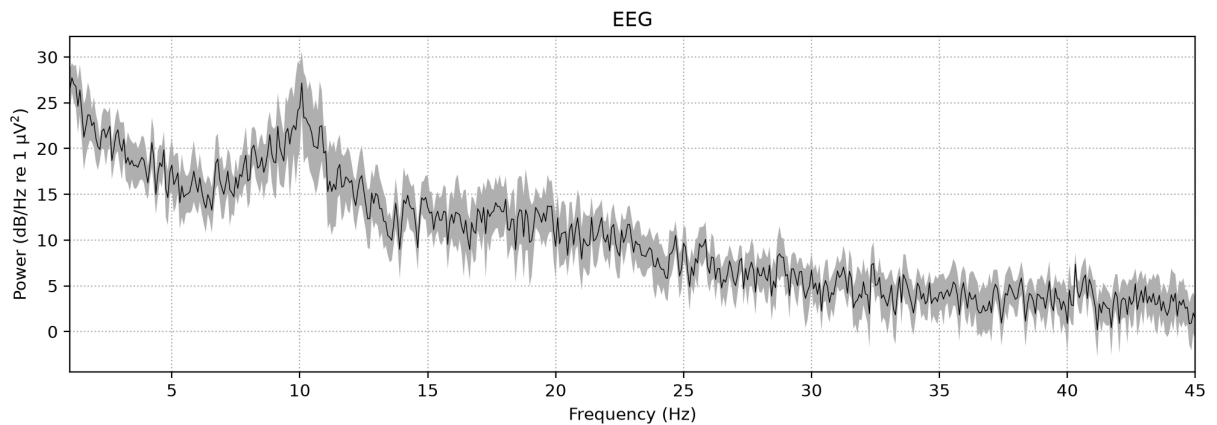
Each value represents estimated spectral power for one EEG channel at one frequency.

Average EEG spectrum

MNE's `Spectrum.plot()` displays spectral power as a function of frequency.

The plot allows us to inspect the overall frequency structure of the EEG recording.

Plotting power spectral density (`dB=True`).



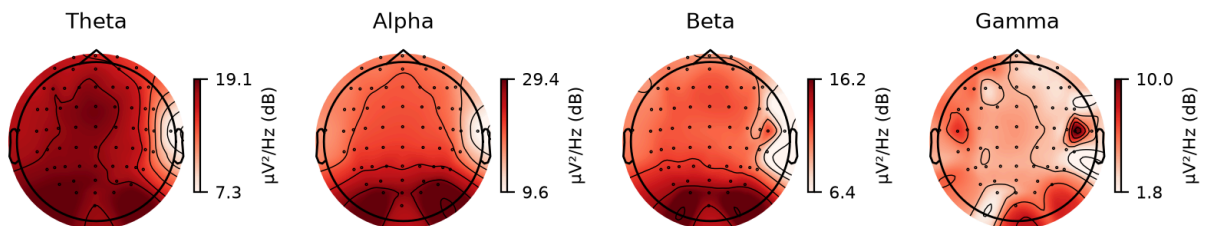
Spatial distribution of spectral power

EEG frequency content can also be represented spatially.

`Spectrum.plot_topomap()` aggregates spectral power inside predefined frequency bands and interpolates the values across the scalp.

We display four common EEG bands:

- **Theta:** 4–8 Hz
- **Alpha:** 8–12 Hz
- **Beta:** 12–30 Hz
- **Gamma:** 30–45 Hz



What this example demonstrates

MNE can represent a continuous EEG recording in the frequency domain using a `Spectrum` object.

From the same object we can obtain:

- numerical PSD values;
- frequency bins;
- an averaged frequency × power visualization;
- spatial scalp maps for selected frequency bands.

This establishes a basic pattern that will be reused in later EEG Feature Atlas examples:

signal → mathematical feature → spatial / spectral visualization

Future examples can extend this workflow to individual channels, epochs, experimental conditions, wavelets, connectivity measures, and features reported in scientific studies.